Network Classes and Subnetting, Troubleshooting

Department of Computer Science and Applications,
MITWPU, Pune

Prepared by:
Dr. Suvarna Sharma

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IPv4 Address

- An IP address is a unique global address for a network interface
- Exceptions:
  - Dynamically assigned IP addresses
  - IP addresses in private networks
- An IP address:

  - is a 32 bit long identifier

  - encodes a network number (network prefix) and a host number

Network Prefix and host number

- The network prefix identifies a network and the host number identifies a specific host (actually, interface on the network).

network prefix	host number

- How do we know how long the network prefix is?
  - Before 1993: The network prefix is implicitly defined (see class-based addressing) or
  - After 1993: The network prefix is indicated by a netmask.

IPv4 ADDRESSES

- An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.
  - Each computer in a TCP/IP network must be given a unique identifier, or IP address.
- This address, which operates at Layer 3, allows one computer to locate another computer on a network.

- An address space is the total number of addresses used by the protocol.
- The address space of IPv4 is 2^{32} or 4,294,967,296 (more than 4 billion).

- If a device operating at the network layer has *m* connections to the internet, it needs to have *m* addresses. E.g.: router

Dotted Decimal Notation

- IP addresses are written in a so-called dotted decimal notation
- Each byte is identified by a decimal number in the range [0..255]:
- Exam

10000000	10001111	10001001	10010000

1st Byte	2nd Byte	3rd Byte	4th Byte
= 128	= 143	= 137	= 144

128.143.137.144

Example

Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78
b. 221.34.7.82

Solution
We replace each decimal number with its binary equivalent.

a. 01101111 00111000 00101101 01001110
b. 11011101 00100010 00000111 01010010

CLASSFUL ADDRESSING

An IP address is 32-bit long. An IP address is divided into sub-classes:

- Class A
- Class B
- Class C
- Class D
- Class E

An IP address is divided into two parts:

- **Network ID:** It represents the number of networks.
- **Host ID:** It represents the number of hosts.

Classful Addressing (contd.)

IN THE DIAGRAM, WE OBSERVE THAT EACH CLASS HAVE A SPECIFIC RANGE OF IP ADDRESSES. THE CLASS OF IP ADDRESS IS USED TO DETERMINE THE NUMBER OF BITS USED IN A CLASS AND NUMBER OF NETWORKS AND HOSTS AVAILABLE IN THE CLASS.

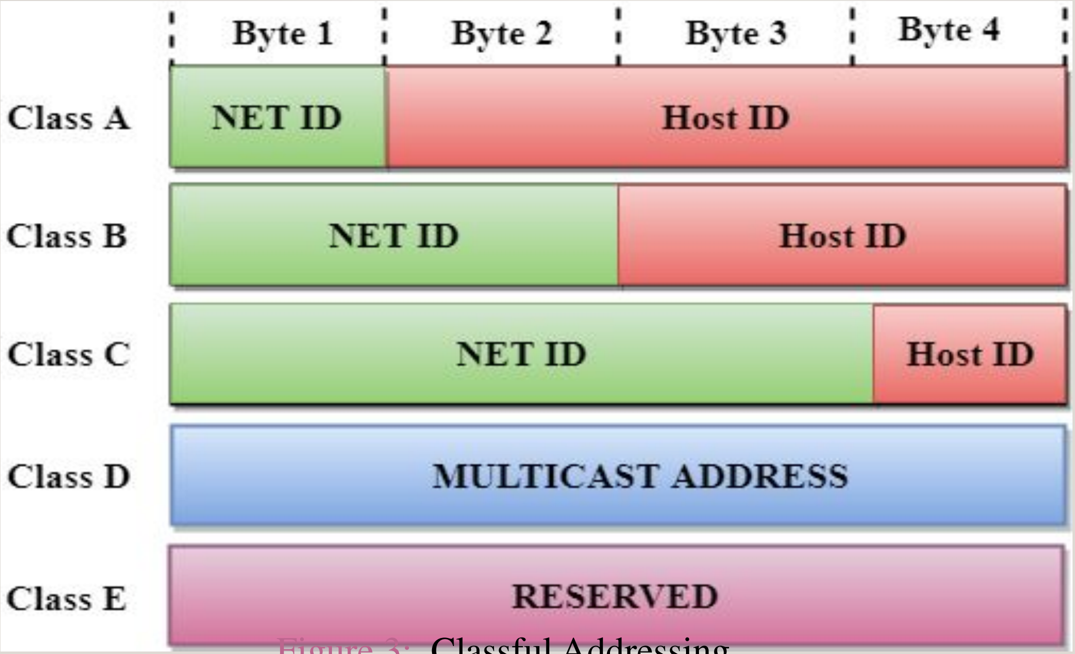


Figure 3: Classful Addressing

Classful Addressing (contd.)

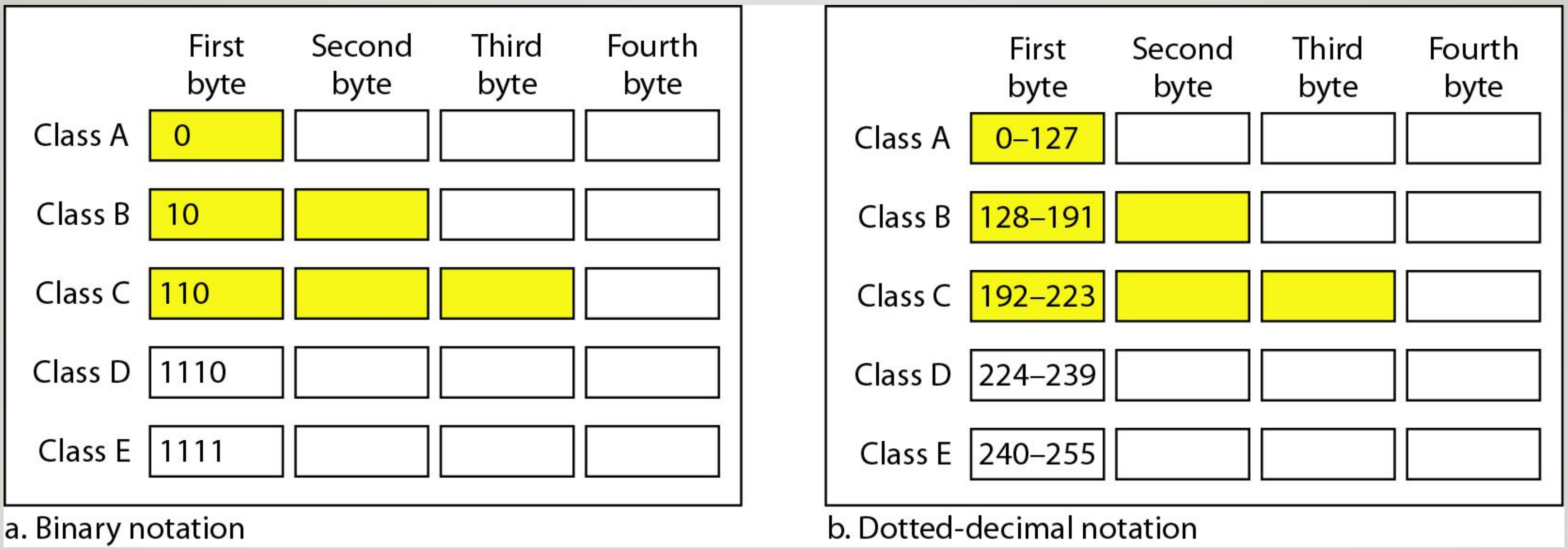


Figure 4: Finding the classes in binary and dotted-decimal notation

Classful Addressing (contd.)

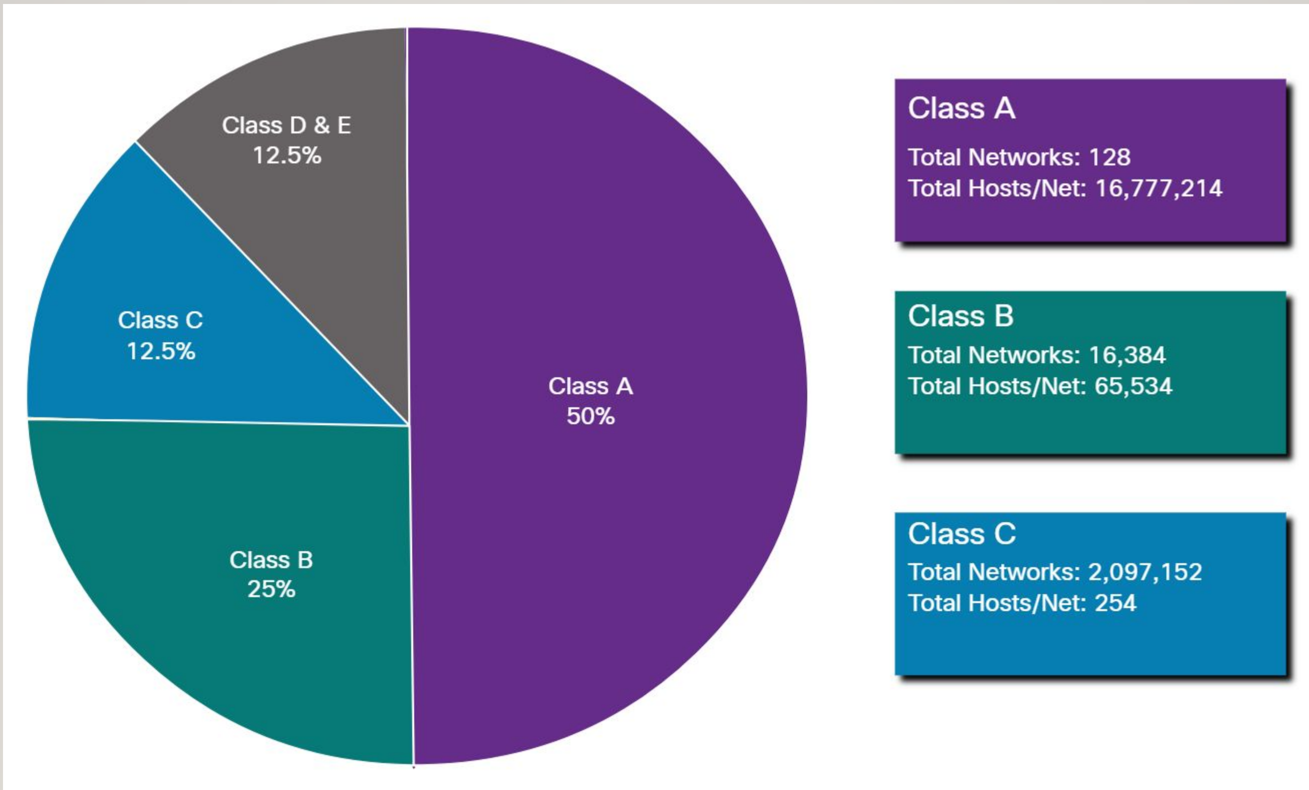


Figure 5: Networks and Hosts in classes

CLASS A

- In Class A, an IP address is assigned to those networks that contain a large number of hosts.
- The network ID is 8 bits long.
- The host ID is 24 bits long.
- In Class A, the first bit in higher order bits of the first octet is always set to 0 and the remaining 7 bits determine the network ID. The 24 bits determine the host ID in any network.
- The total number of networks in Class A = 2^7 = 128 network address
- The total number of hosts in Class A = 2^{24} - 2 = 16,777,214 host address

CLASS B

- In Class B, an IP address is assigned to those networks that range from small-sized to large-sized networks.
- The Network ID is 16 bits long.
- The Host ID is 16 bits long.
- In Class B, the higher order bits of the first octet is always set to 10, and the remaining14 bits determine the network ID. The other 16 bits determine the Host ID.
- The total number of networks in Class B = 2^{14} = 16384 network address
- The total number of hosts in Class B = $2^{16} - 2$ = 65534 host address

CLASS C

- In Class C, an IP address is assigned to only small-sized networks.
- The Network ID is 24 bits long.
- The host ID is 8 bits long.
- In Class C, the higher order bits of the first octet is always set to 110, and the remaining 21 bits determine the network ID. The 8 bits of the host ID determine the host in a network.
- The total number of networks = 2^{21} = 2097152 network address
- The total number of hosts = 2^8 - 2 = 254 host address

CLASS D

- In Class D, an IP address is reserved for multicast addresses. It does not possess subnetting.
- The higher order bits of the first octet is always set to 1110, and the remaining bits determines the host ID in any network.

CLASS E

- In Class E, an IP address is used for the future use or for the research and development purposes. It does not possess any subnetting. The higher order bits of the first octet is always set to 1111, and the remaining bits determines the host ID in any network.

Classful Addressing

- A block in class A address is too large for almost any organization. It means that most of the IP addresses in class A were wasted.

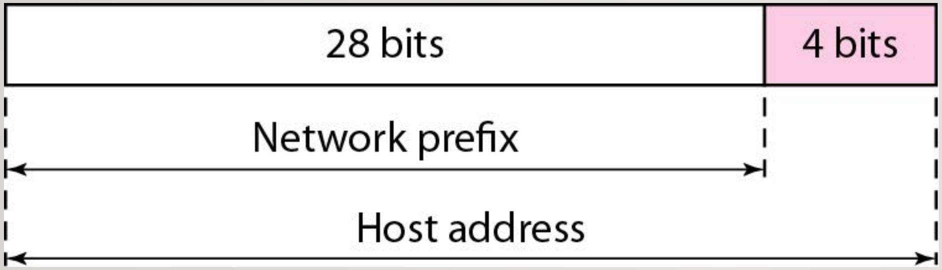
- A block in class B is also very large.

- A block in class C is probably too small for many organizations.

- Class D addresses were designed for multicasting. Each IP address in this class is used to define one group of hosts on the Internet. The Internet authorities wrongly predicted a need of 268,435,456 groups. This never happened and many IP addresses were wasted.

- Class E IP addresses were reserved for future use; only a few were used, resulting in another waste of IP addresses

Two level hierarchy in an IPv4 address

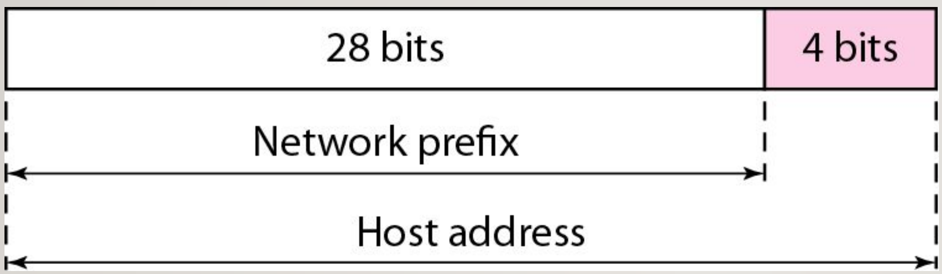


Leftmost n bits (prefix) define the network id and,

The right most 32-n (suffix) bits define the host id.

The prefix is common to all devices in the network, suffix changes from one device to another.

Two level hierarchy in an IPv4 address



Leftmost n bits (prefix) define the network id and,

The right most 32-n (suffix) bits define the host id.

The prefix is common to all devices in the network, suffix changes from one device to another.

Netid and Hostid:

- This concept does not apply to class D and E.
- For class A, one byte defines netid, and three bytes define hostid.
- In class B, two bytes define the netid, and two bytes define the hostid.
- In class C, three bytes define the netid , and one byte defines the hostid.

Network Masks

Mask:

- The concept does not apply to classes D and E.
- The mask can help us find netid and hostid.
- The last column shows the mask in the form /n, where n can be 8,16, or 24. This notation is called slash notation or Classless Interdomain Routing (CIDR) notation.

Class	*Binary*	*Dotted-Decimal*	*CIDR*
A	**11111111** 00000000 00000000 00000000	**255**.0.0.0	/8
B	**11111111 11111111** 00000000 00000000	**255.255**.0.0	/16
C	**11111111 11111111 11111111** 00000000	**255.255.255**.0	/24

Figure 6: Default masks for classful addressing

Network Masks (Contd.)

An IP address on a Class A network that has not been subnetted would have an address/mask pair similar to: 10.20.15.1 255.0.0.0.
In order to see how the mask helps you identify the network and node parts of the address, convert the address and mask to binary numbers.

10.20.15.1 = 00001010.00010100.00001111.00000001
255.0.0.0   = 11111111.00000000.00000000.00000000

Once you have the address and the mask represented in binary, then identification of the network and host ID is easier. Any address bits which have corresponding mask bits set to 1 represent the network ID. Any address bits that have corresponding mask bits set to 0 represent the node ID.

10.20.15.1 = 00001010.00010100.00001111.00000001
255.0.0.0   = 11111111.00000000.00000000.00000000
----
net id | host id

netid = 00001010 = 10
hostid = 00010100.00001111.00000001 = 20.15.1

RULES FOR ASSIGNING HOST ID:

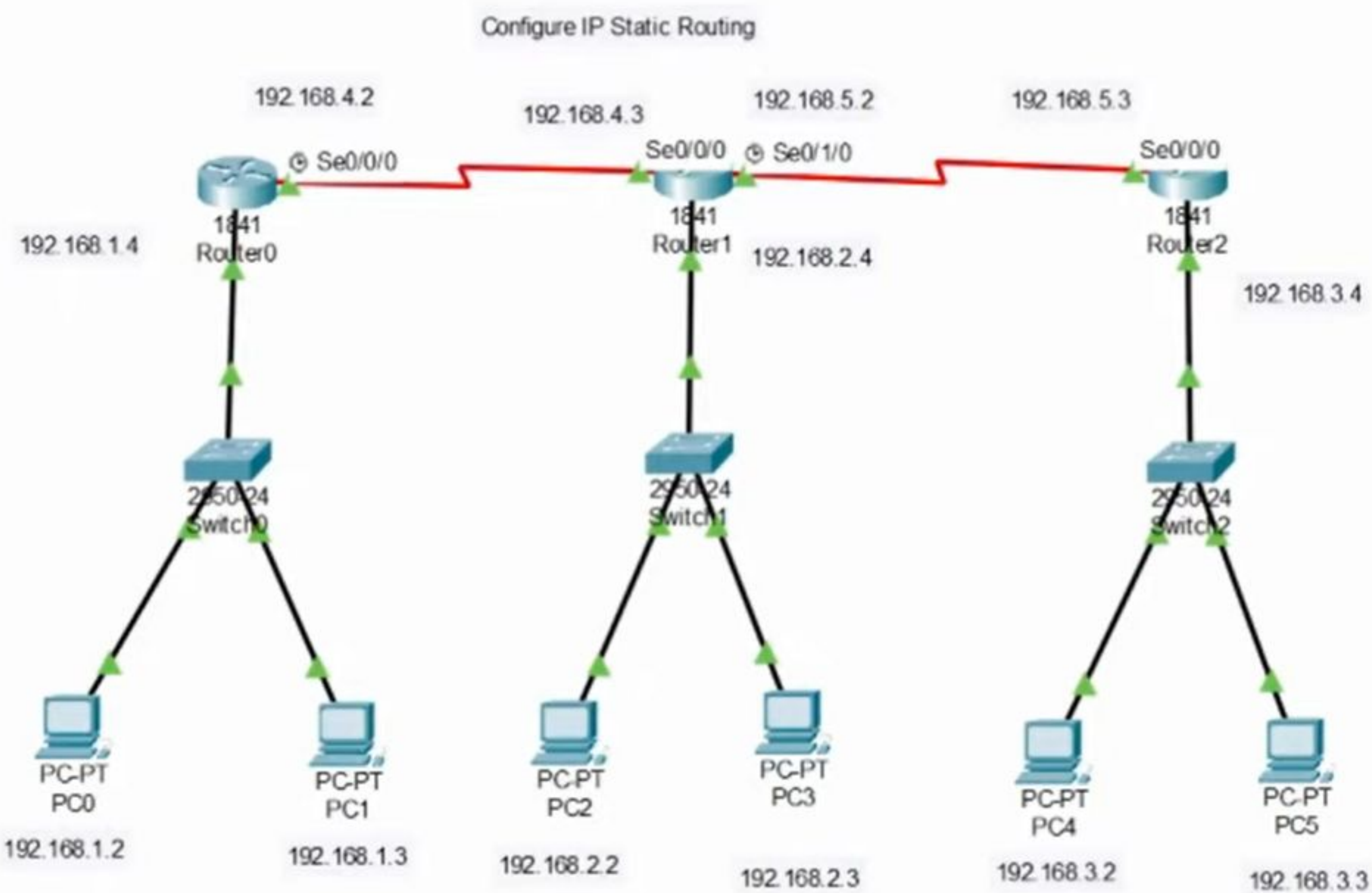
The Host ID is used to determine the host within any network. The Host ID is assigned based on the following rules:

- The Host ID must be unique within any network.
- The Host ID in which all the bits are set to 0 cannot be assigned as it is used to represent the network ID of the IP address.
- The Host ID in which all the bits are set to 1 cannot be assigned as it is reserved for the multicast address.

RULES FOR ASSIGNING NETWORK ID:

If the hosts are located within the same local network, then they are assigned with the same network ID. The following are the rules for assigning Network ID:

- The network ID cannot start with 127 as 127 is used by Class A.
- The Network ID in which all the bits are set to 0 cannot be assigned as it is used to specify a particular host on the local network.
- The Network ID in which all the bits are set to 1 cannot be assigned as it is reserved for the multicast address.



Configure IP Static Routing
192.168.4.2
192.168.4.3
192.168.5.2
192.168.5.3
Se0/0/0
Se0/0/0
Se0/1/0
Se0/0/0
1841
Router0
1841
Router1
1841
Router2
192.168.1.4
192.168.2.4
192.168.3.4
2950-24
Switch0
2950-24
Switch1
2950-24
Switch2
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
192.168.1.2
192.168.1.3
192.168.2.2
192.168.2.3
192.168.3.2
192.168.3.3

Classful Addressing Chart

Class	*Number of Blocks*	*Block Size*	*Application*
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Figure 7: Number of blocks and block size in classful IPv4 addressing

Classful Addressing Chart

Class	Higher bits	NET ID bits	HOST ID bits	No. of networks	No. of hosts per network	Range
A	0	8	24	2^7	2^{24}	0.0.0.0 to 127.255.255.255
B	10	16	16	2^{14}	2^{16}	128.0.0.0 to 191.255.255.255
C	110	24	8	2^{21}	2^8	192.0.0.0 to 223.255.255.255
D	1110	Not defined	Not defined	Not defined	Not defined	224.0.0.0 to 239.255.255.255
E	1111	Not defined	Not defined	Not defined	Not defined	240.0.0.0 to 255.255.255.255

Classless Addressing

Classless Addressing is one of the IP(Internet Protocol) address classifications. IP addresses are unique and universal. No two hosts will simultaneously have the same IP address in the world. The most common use of the classless addressing system Or Classless interdomain routing (CIDR) for actually addressing is to combine two or more class C networks to create a /23 or /22 Supernet.

To overcome address depletion and give Internet access to more organizations, classless addressing was designed and implemented.

In this scheme, there are no classes but the addresses are still granted in blocks.

Number of Addresses in a Block

There is only one condition on the number of addresses in a block; it must be a power of 2 (2, 4, 8, ...).

- A household may be given a block of 2 addresses.
- A small business may be given 16 addresses.
- A large organization may be given 1024 addresses.

Classless Addressing (Contd.)

Beginning Address

The beginning address must be evenly divisible by the number of addresses.

- For example, if a block contains 4 addresses, the beginning address must be divisible by 4.
- If the block has less than 256 addresses, we need to check only the rightmost byte.
- If it has less than 65,536 addresses, we need to check only the two rightmost bytes, and so on.

Examples

Which of the following can be the beginning address of a block that contains 16 addresses?

205.16.37.32
190.16.42.44
17.17.33.80
123.45.24.52

$16 = 2^4 \rightarrow$ **last 4 bits = 0**
Last 4 bits are part of the **last octet**
So we check:
Is the **last octet divisible by 16?**
That’s why we divide by **16**

Solution
The address 205.16.37.32 is eligible because 32 is divisible by16.
The address 17.17.33.80 is eligible because 80 is divisible by 16.

Examples

Which of the following can be the beginning address of a block that contains 1024 addresses?

205.16.37.32
190.16.42.0
17.17.32.0
123.45.24.52

Solution

A block of 1024 addresses = 2^{10}, so the starting (beginning) address must have its last 10 bits = 0.

This means:

The block size is 1024 → subnet mask is /22 (since 32 − 10 = 22)

Valid starting addresses must be multiples of 1024 in terms of address numbering

Quick rule:

Look at the last two octets

Since 1024 = 4 × 256, the third octet must be a multiple of 4, and the fourth octet must be 0

Subnetting

- If an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks (called subnets).

- Subnetting increases the number of 1s in the mask.

Supernetting

- The size of a class C block with a maximum number of 256 addresses did not satisfy the needs of most organizations.

- In supernetting, an organization can combine several class C blocks to create a larger range of addresses.

- In other words, several networks are combined to create a super network or supernet.
- E.g. an organization that needs 1000 addresses can be granted four contiguous class C blocks.

- Supernetting decreases the number of 1s in the mask.
- E.g. if an organization is given four class C blocks, the mask changes from /24 to /22.
- Classless addressing eliminated the need for supernetting.

Address Blocks

In classless addressing, when an entity small or large, needs to be connected to the Internet, it is granted a block of addresses. The size of the block varies based on the need of the organization.

Restriction-

1. The addresses in a block must be contiguous.
2. The no of addresses in a block must be a power of 2.
3. The first address must be evenly divisible by the number of addresses.

Example

Figure (on next slide) shows a block of addresses, in both binary and dotted-decimal notation, granted to a small business that needs 16 addresses.

We can see that the restrictions are applied to this block. The addresses are contiguous.

The number of addresses is a power of 2 ($16 = 2^4$), and the first address is divisible by 16.

The first address, when converted to a decimal number, is 3,440,387,360, which when divided by 16 results in 215,024,210.

Example (contd.)

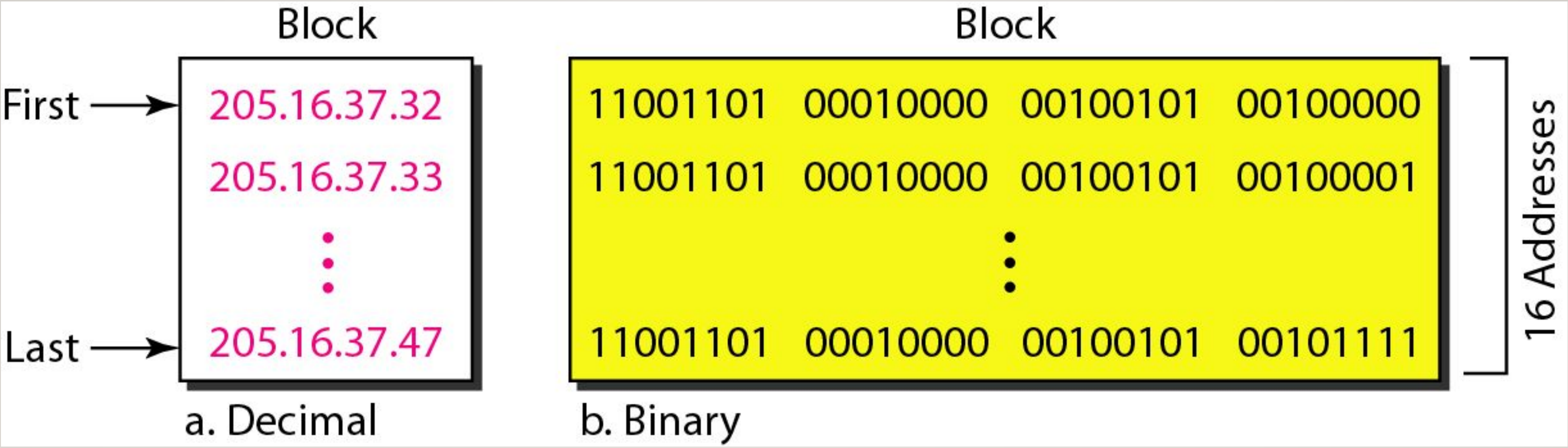


Figure 8: A block of 16 addresses granted to a small organization

In IPv4 addressing, a block of addresses can be defined as

x.y.z.t /*n*

in which x.y.z.t defines one of the addresses and the /*n* defines the mask.

The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.

Example

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Example

Solution:

The block of addresses 205.16.37.39/28 represents a subnet with a subnet mask of 28 bits. To determine the first, we need to convert the subnet mask to its binary representation and perform some calculations.

The subnet mask 28 means that the first 28 bits of the IP address represent the network portion, and the remaining 4 bits are available for host addresses.

Converting the subnet mask 28 to binary results in 32 bits with the first 28 bits set to 1 and the last 4 bits set to 0:

11111111.11111111.11111111.11110000

To calculate the first address, we perform a logical AND operation between the given address (205.16.37.39) and the subnet mask:

Address: 11001101.00010000.00100101.00100111
Subnet Mask: 11111111.11111111.11111111.11110000
AND Operation: 11001101.00010000.00100101.00100000

The result of the AND operation gives us the network address, which is 205.16.37.32. Therefore, the first address in the block is 205.16.37.32.

Network Address:

- The first address is called the network address and defines the organization network
- First address is used as the network address that represents the organization to the rest of the world.

The last address in the block can be found by setting the rightmost 32 - n bits to 1s.

Example

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the last address in the block?

Example

205.16.37.39/28

Solution:

To calculate the last address, we need to find the highest possible host address within the block. Since we have 4 bits available for host addresses (2^4 = 16), the highest host address is obtained by setting all 4 bits to 1:

Address: 11001101.00010000.00100101.00100111
Subnet Mask: 11111111.11111111.11111111.11110000
Subnet Mask Complement: 00000000.00000000.00000000.00001111

	11001101.00010000.00100101.00100111
	00000000.00000000.00000000.00001111
OR Operation:	11001101.00010000.00100101.00101111

The result of the OR operation, with the complement of the mask, gives us the broadcast address, which is 205.16.37.47. However, the last address in the block is always reserved as the broadcast address, so the actual last usable address is the one before the broadcast address. Therefore, the last address in the block is 205.16.37.46.

The number of addresses in the block can be found by using the formula 2^{32-n}.

Three level of Hierarchy: subnetting

An organization that is granted a large block of addresses may want to create clusters of networks (subnets) and divide the addresses between the different subnets.

The rest of the world still sees the organization as one entity; however internally there are several subnets.

All messages are sent to the router that connects the organization to the rest of the Internet; the router routes the message to the appropriate subnets.

The organization has its own mask and each subnet must also have its own.

Example

Suppose an organization is given the block 17.12.40.0/26, which contains 64 IP addresses. The organization has three offices and needs to divide the addresses into three subblocks of 32,16, and 16 IP addresses.

We can find the new masks as

1. *Suppose the mask for the first subnet is n_1 then 2^{32-n1} must be 32, which means n_1=27.*
2. *Suppose the mask for the second subnet is n_2 then 2^{32-n2} must be 16, which means n_1=28.*
3. *Suppose the mask for the third subnet is n_3 then 2^{32-n3} must be 16, which means n_1=28.*

Example (contd.)

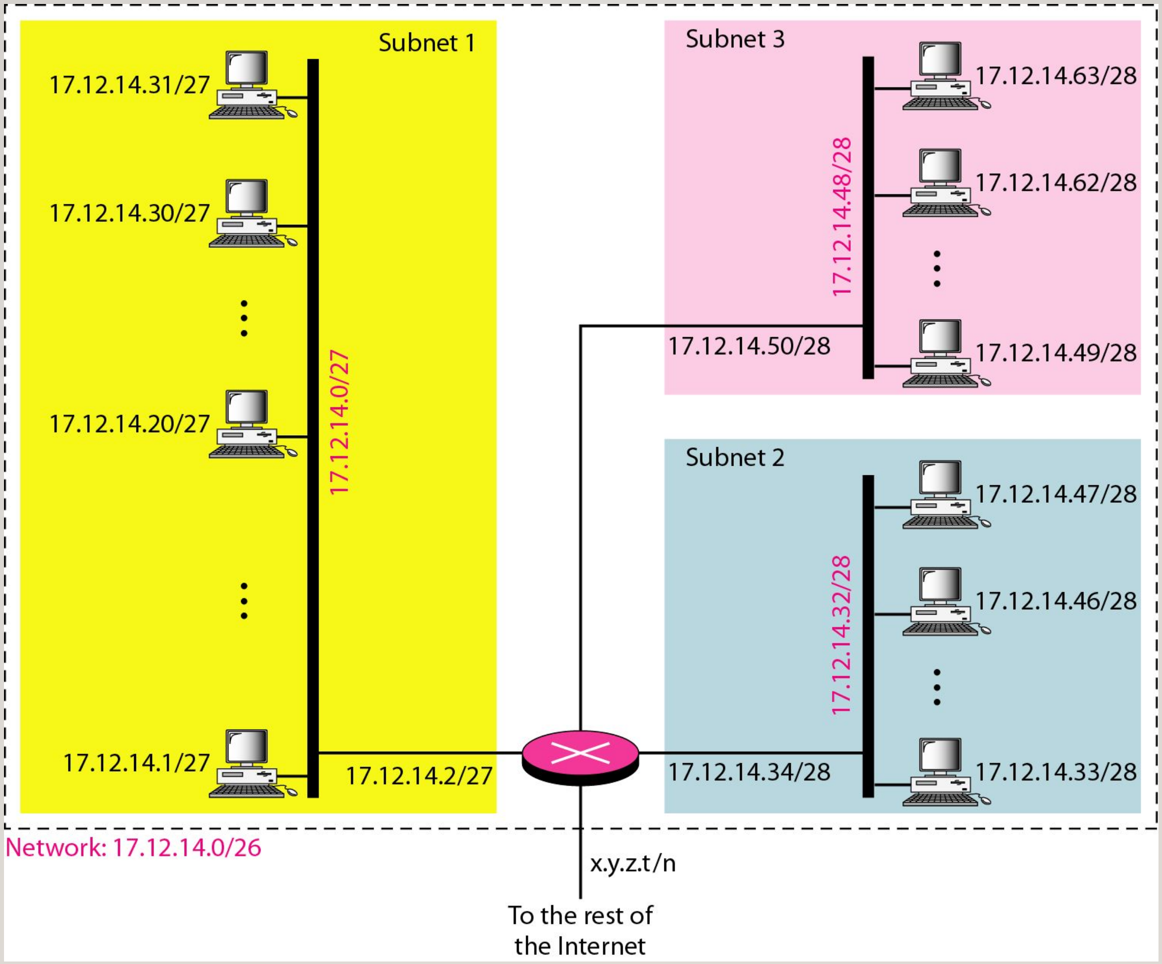


Figure 9: Configuration and addresses in a subnetted network

Three-level hierarchy in an IPv4 address

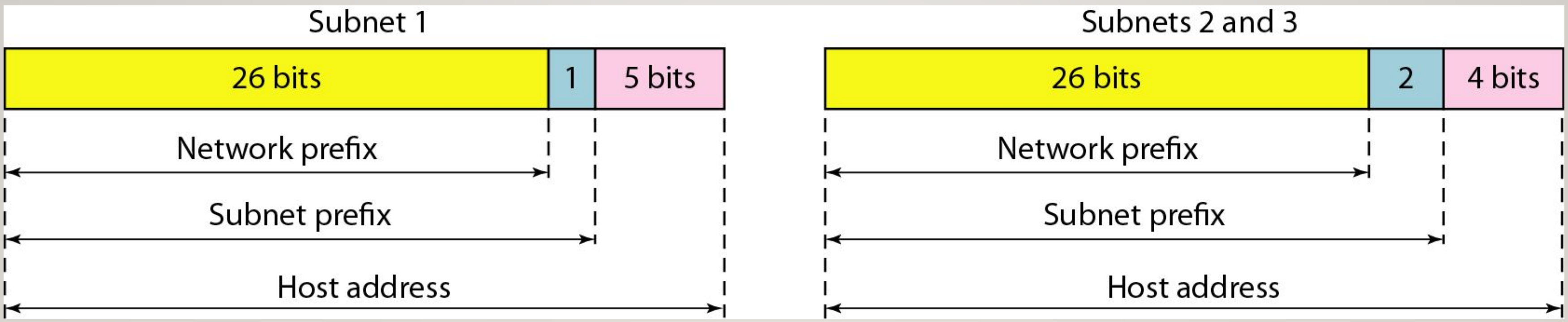


Figure 10: *Three-level hierarchy in an IPv4 address*

Example

Given IP address 182.168.1.0/20, calculate: • Number of subnets • Number of hosts per subnet • Valid host range for the first subnet

1) Number of subnets

- Default mask for Class B = **/16**
- Given mask = **/20**
- Borrowed bits = 20 − 16 = **4 bits**
- Number of subnets = $2^4 = 16$

2) Number of hosts per subnet

Host bits = 32 − 20 = **12 bits**

Total addresses per subnet = $2^{12} = 4096$

Usable hosts = 4096 − 2

3) Valid host range for the first subnet

Block size = **16** in the third octet (because /20)

Subnets start from:

- 182.168.0.0
- 182.168.16.0
- 182.168.32.0 …

First subnet:

Network address: **182.168.0.0**

Broadcast address: **182.168.15.255**

Valid host range: **182.168.0.1 to 182.168.15.254**

Example

Given IP address 192.16.0.0/26, calculate: • Number of subnets • Number of hosts per subnet • Subnet mask in dotted decimal form

Default Class of Address: Class C   subnet mask 24 bits

Number of subnets

bits in subnet mask 26

Borrowed bits = 26 − 24 = 2

Number of subnets = $2^2 = 4$

Number of hosts per subnet

Host bits = 32 − 26 = 6

Total hosts = $2^6 = 64$

Usable hosts = 64 − 2 = 62

Subnet mask in dotted decimal form

11111111.11111111.11111111.11000000

Subnet mask = 255.255.255.192

Example

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

a. The first group has 64 customers; each needs 256 addresses.
b. The second group has 128 customers; each needs 128 addresses.
c. The third group has 128 customers; each needs 64 addresses.

Design the subblocks and find out how many addresses are still available after these allocations.

Example (contd.)

Solution

Group 1

For this group, each customer needs 256 addresses. This means that 8 ($\log_2$ 256) bits are needed to define each host. The prefix length is then 32 − 8 = 24. The addresses are

1st Customer:	*190.100.0.0/24*	*190.100.0.255/24*
2nd Customer:	*190.100.1.0/24*	*190.100.1.255/24*
. . .		
64th Customer:	*190.100.63.0/24*	*190.100.63.255/24*

Total = 64 × 256 = 16,384

Example (contd.)

Group 2

For this group, each customer needs 128 addresses. This means that 7 ($\log_2$ 128) bits are needed to define each host. The prefix length is then 32 − 7 = 25. The addresses are

1st Customer:	*190.100.64.0/25*	*190.100.64.127/25*
2nd Customer:	*190.100.64.128/25*	*190.100.64.255/25*
...		
128th Customer:	*190.100.127.128/25*	*190.100.127.255/25*

Total = 128 × 128 = 16,384

Example (contd.)

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then 32 − 6 = 26. The addresses are

1st Customer:	*190.100.128.0/26*	*190.100.128.63/26*
2nd Customer:	*190.100.128.64/26*	*190.100.128.127/26*
. . .		
128th Customer:	*190.100.159.192/26*	*190.100.159.255/26*

Total $= 128 \times 64 = 8192$

Number of granted addresses to the ISP: 65,536
Number of allocated addresses by the ISP: 40,960
Number of available addresses: 24,576

Example (contd.)

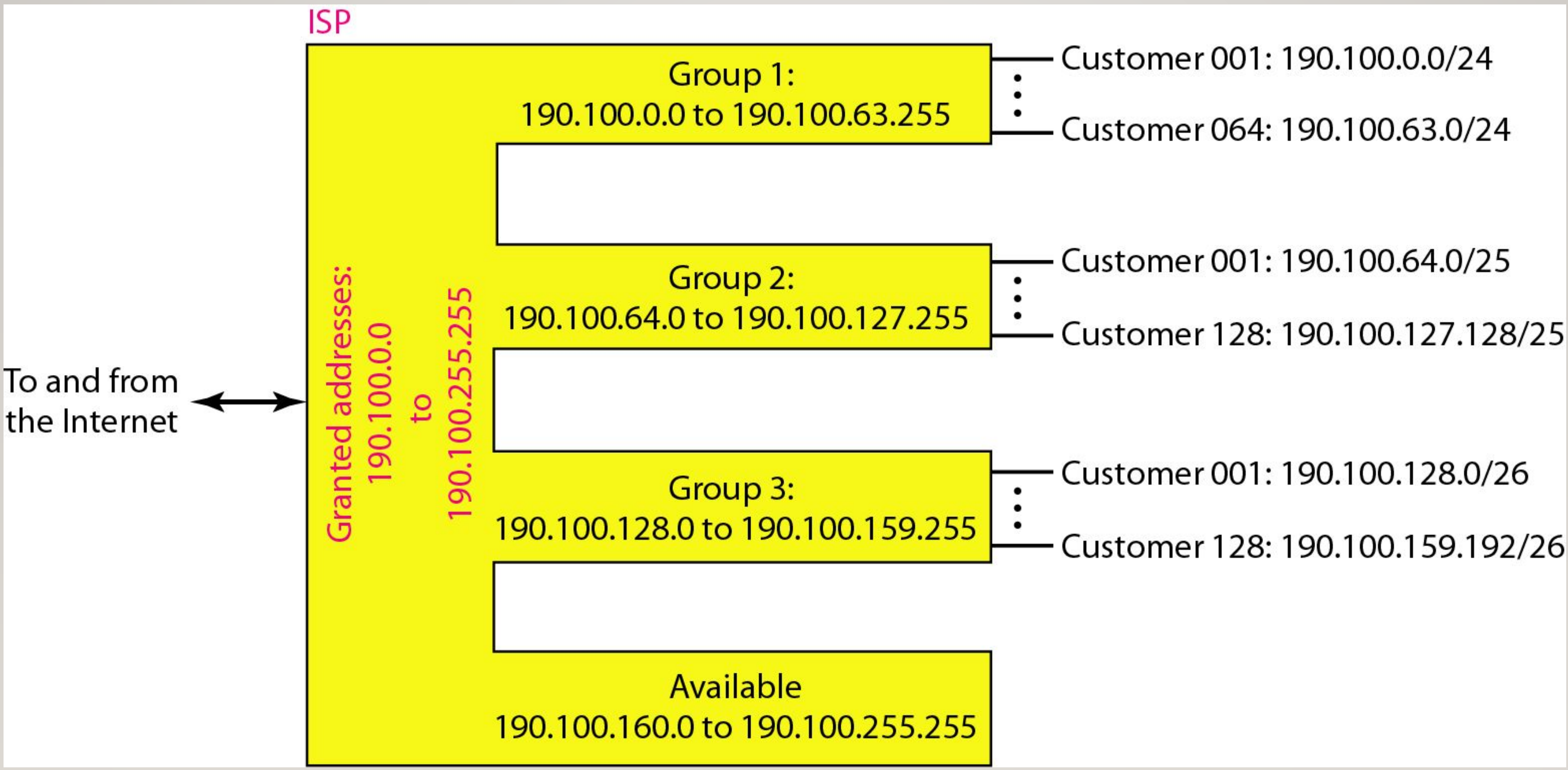


Figure 11: An example of address allocation and distribution by an ISP

Practice Questions

Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11000001 10000011 00011011 11111111

Practice Questions

Find the error, if any, in the following IPv4 addresses.

a. 111.56.045.78
b. 221.34.7.8.20
c. 75.45.301.14
d. 11100010.23.14.67

Practice Questions

Find the class of each address.

a. 00000001 00001011 00001011 11101111
b. 11000001 10000011 00011011 11111111
c. 14.23.120.8
d. 252.5.15.111

Practice Questions

An organization is facing IP address exhaustion and plans to redesign its network.
For the Given IP address: 180.16.37.39/20, calculate:

a) Address Mask
b) First address of the block
c) Last address of the block
d) Number of addresses in the block

Practice Questions

The block of addresses 205.16.37.39/28 represents a subnet with a subnet mask of 28 bits.

To determine the first and last addresses in the block, we need to convert the subnet mask to its binary representation and perform some calculations.

The subnet mask 28 means that the first 28 bits of the IP address represent the network portion, and the remaining 4 bits are available for host addresses.

Converting the subnet mask 28 to binary results in 32 bits with the first 28 bits set to 1 and the last 4 bits set to 0:

11111111.11111111.11111111.11110000

To calculate the first address, we perform a logical AND operation between the given address (205.16.37.39) and the subnet mask:

Address: 11001101.00010000.00100101.00100111

Subnet Mask: 11111111.11111111.11111111.11110000

AND Operation: 11001101.00010000.00100101.00100000

The result of the AND operation gives us the network address, which is 205.16.37.32. Therefore, the first address in the block is 205.16.37.32.

Practice Questions

To calculate the last address, we need to find the highest possible host address within the block.
Since we have 4 bits available for host addresses (2^4 = 16), the highest host address is obtained by setting all 4 bits to 1:

Address: 11001101.00010000.00100101.00100111
Subnet Mask: 00000000.00000000.00000000.00001111
OR Operation: 11001101.00010000.00100101.00101111

The result of the OR operation gives us the broadcast address, which is 205.16.37.47. However, the last address in the block is always reserved as the broadcast address, so the actual last usable address is the one before the broadcast address. Therefore, the last address in the block is 205.16.37.46.

The total number of addresses in the block can be calculated by taking the number of available host addresses, which is 2^4 = 16, and subtracting 2 (one for the network address and one for the broadcast address). Therefore, the total number of addresses in the block is 16 - 2 = **14**.

happy

LEARNING